



Day 1: Python Fundamentals

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Topics Covered:

1. Virtual Environment
2. VSCode / Jupyter Notebook / Kaggle
3. Commands
4. Data Types
5. Dictionary
6. List
7. Indexing
8. Slicing
9. Conditional Statements
10. For Loop
11. While Loop

Data Types

```
# DataTypes
x = int(input('Enter a random number:'))
print(type(x))

# string to int
```

```
x = '12'  
print(int(x))
```

Dictionary

```
dict = {'x':'gaurav', 'y':'aditya', 'z':'sakshiee'}  
print(dict['y'])
```

List

```
list = [2, 'gaurav', 3.14, True, dict]  
print(list)
```

Indexing

```
print(list[3])    # +ve indexing  
print(list[-2])   # -ve indexing
```

Slicing

```
print(list[1:3])
```

String Practice Questions

```
str = 'my name is Gaurav, im from sapkal'  
  
# Q2: Print characters up to index 6  
print(str[0:6])  
  
# Q3: Print string by removing first 2 chars  
print(str[2:])
```

```
# Q4: Print middle char from str
print(str[11:17])
```

Conditional Statements

```
marks = int(input('Enter your marks:'))

if marks <= 80:
    print("Grade C")
elif marks > 80 and marks <= 90:
    print("Grade B")
else:
    print("Grade A")
```

For Loop

```
# Example 1
for i in range(1, 11):
    print(i)

# Example 2
for i in range(0, 21, 2):
    print(i)
```

While Loop

```
# Example 1: Multiplication table
n = int(input("Enter a number:"))
i = 1
while i <= 10:
    print(i * n)
    i += 1
```

```
# Example 2: Search in list
list = [5, 10, 15, 20, 25, 30, 35, 40, 45, 50]
num = int(input("Enter number to find:"))
i = 0
while i < len(list):
    if list[i] == num:
        print('Number found at', i)
    i += 1
```